Problem Set 2

Due date: October 14th, 17:00.

Please upload all answers to Canvas. This includes pdf scans of whatever hand written solutions you produce, any code which you write (including instructions on how to execute that code) and any outputs that code produces. For all of your code please be sure it is fully commented, and for any plots be sure that axes have labels and are legible.

Problem 1 - The Ising Model

In the Ising model, a grid of Ising spins (particles which can have a spin of $s_i = \pm 1$) are arranged in a grid. Each particle is allowed to interact with its neighbour via an exchange energy given by

$$E_i = -2J \sum_{<ij>} s_i s_j$$

The magnetisation per spin of such a grid with N spins is

$$M = \frac{1}{N} \sum_i s_i$$

Note that in the following I have assumed dimensionless units for all variables (T,H).

1. Following the procedure in the notes, write a program which computes the magnetisation per spin of a square grid of N Ising spins for a given temperature, where each spin is randomly assigned a spin in the first instance. Using this program, create a plot of the magnetisation versus Monte Carlo time for: T=1, T=2, T=2.22, T=2.29, T=3, T=5. You should run the code for a minimum of 1000 Monte Carlo times. **(5 marks)**
2. By time-averaging the magnetisation, the total energy per spin of the system, and the heat capacity per spin for a given temperature, create a plot of the magnetisation, total energy, and heat capacity of the system as a function of temperature. I suggest using the temperature range of T=range(1,5,0.03). The heat capacity is given by

   $$C = \frac{(\Delta E)^2}{kT^2}$$

   where the variance in the energy is givem by $(\Delta E)^2 = \left\langle E^2 \right\rangle - \left\langle E \right\rangle^2$ and where $\langle\rangle$ represents the time average of a value. **(5 marks)**
3. Imagine the system is exposed to an external magnetic field H which either points up or down. This changes the energy of a given spin to

   $$E_i = -2J \sum_{<ij>} s_i s_j - H s_i$$

   where H has been re-normalised. As shown in the lectures, this leads to the

identification of hysteresis effects when the temperature of the bath is low as the external field is either swept from low to high values or high to low. Create a plot of M versus H for $T = 0.25$ and $T = 5$, showing that this leads to hysteresis effects below the critical temperature but not above it. Note that when sweeping from negative H to positive H you'll need to start with a spin lattice where all spins are pointed down, and when sweeping from positive to negative you'll need to start with a lattice where all spins are pointed up. **(5 marks)**